Course One
Foundations of Data Science



Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- ☑ ~~Complete the questions in the Course 1 PACE strategy document~~
- ☑ ~~Answer the questions in the Jupyter notebook project file~~
- ☑ ~~Complete coding prep work on project's Jupyter notebook~~
- ☑ ~~Summarize the column Dtypes~~
- ☑ ~~Communicate important findings in the form of an executive summary~~

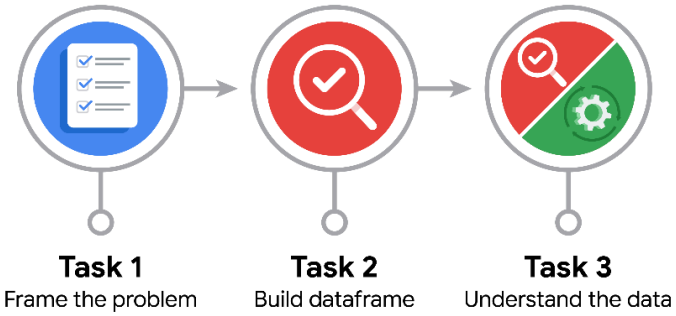
Relevant Interview Questions

Completing the end-of-course project will help you respond these types of questions that are often asked during the interview process:

- Describe the steps you would take to clean and transform an unstructured data set.
- What specific things might you look for as part of your cleaning process?
- What are some of the outliers, anomalies, or unusual things you might look for in the data cleaning process that might impact analyses or ability to create insights?

Reference Guide

This project has three tasks; the visual below identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- How can you best prepare to understand and organize the provided information?

Break down the information to understand smaller questions of the overall ask. For example, who is this for? Waze. What language would be best to use here? Python. What are my initial steps? Understand the ask, understand the data, and understand the variables.

- What follow-along and self-review codebooks will help you perform this work?

The course 1 end-of-course Jupyter notebook.

- What are some additional activities a resourceful learner would perform before starting to code?

Research and planning.



PACE: Analyze Stage

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?

Yes, the dataset has the information required to find the summary of the data, the data types of the data, where missing values are, and descriptive statistics.

- How would you build summary dataframe statistics and assess the min and max range of the data?

First, I would import pandas, read the CSV as a dataframe AS df, and then, for a summary of the statistics, I can run df.describe()

- Do the averages of any of the data variables look unusual? Can you describe the interval data?

The mean for most columns appears to be closer to the upper quartile than the lower quartile, implying that the dataset may include “power users” that increase the mean. However, just looking at the summary statistics, I don’t see any average that is unusual.



PACE: Construct Stage

Note: The Construct stage does not apply to this workflow. The PACE framework can be adapted to fit the specific requirements of any project.



PACE: Execute Stage

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing exploratory data analysis?

1. How and when they consider a user “churned”

2. Why the label column has missing values

3. If the data is representative of all users

4. If the data is logically consistent

- What data initially presents as containing anomalies?

When looking at the max column, we can notice that there is a very heavy power user. Although it could be something like a very dedicated long-haul trucker, it could be possible that this could be something like a shared company vehicle that has a phone in it that only uses Waze. We should consider the reliability of using data from accounts like this.

- What additional types of data could strengthen this dataset?

Km_per_drive, km_per_driving_day, and drives_per_driving_day all make it easier to analyze how much users are driving on a daily basis.
More nominal and True/False rows could also be useful, like “Is this a company account?” just as an example.